

Software Test Document

for

Library Automate

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1. Introduction

This document contains the test plan for Library Automate project of Trinidad Municipal College. This contains the features to be tested, procedures for executing the defined tests, and identifies specific test cases. In test cases, required inputs and expected results are recognized, procedures are provided and outlines the pass/fail criteria for determining acceptance.

1.1. System Overview

Library Automate is a web-based system intended for Trinidad Municipal College's school library that performs computerized library operations and generates reports necessary for monitoring library usage. The system will help the library in keeping borrower's and book's information as well as monitoring the library users and borrowed items.

1.2. Test Approach

The testing approach for the Library Automate project will carry out the Unit, Integration and Acceptance test levels. The developer will carry out the Unit Testing and will be approved by the project adviser. Integration testing will also be performed by the developer. Acceptance testing will be done by the school librarian with the assistance of the developer. The acceptance test will be in parallel with the existing manual process. Proof of testing will be provided to the adviser for complete acceptance of each module.

1.3. Definition, Acronyms, and Abbreviations

Term	Definition
TMC	Trinidad Municipal College
LGU	Local Government Unit
MVC Framework	Model-View-Controller Framework. Is a software architectural pattern commonly used for developing user interfaces which divides the related program logic into three interconnected elements.
Test Case	Test Case. Is a set of conditions or variables under which a tester will determine whether a system under test satisfies requirements or works correctly.
Test Plan	Test Plan. Is a document describing software testing scope and activities.

Table 1.0 Definition, Acronyms, and Abbreviations

2. Test Plan

This section lists the things the proponent intends to test within the scope of this test plan. The features to be tested are intended to be delivered to the client. It is assumed that unit testing will be done through black box testing. The modules and sub modules to be tested in the Library Automate are as follows:

Module	Sub Module
Login	• Librarian Personnel (Librarian, Asst. Librarian, and Staff) • Faculty • Students
Borrowers Information Management	• Add Borrower's Information • Update Borrower's Information • Deactivate Borrower's Information
Book Information Management	• Add Book's Information • Update Book's Information • Deactivate Book's Information
Book Issuance	
Return Book	
Report	• Transactions Report • Library Visitors • Borrowers • Items • Inventory

Table 2.0 Modules and Sub Modules

2.1. Features to be Tested

System Features	Priority	Task ID	Description
Login			
User Login	H	TID 1.0	Library personnel, faculty and students access roles will be tested for login
Borrowers Information Management			
Add Borrower's Information	H	TID 2.1	Adds new borrower's information into the

			database
Update Borrower's Information	H	TID 2.2	Updates borrower's information from the database
Deactivate Borrower's Record	H	TID 2.3	Updates borrower's status to mark it as deactivated record from database
Book's Information Management			
Add Book's Information	H	TID 3.1	Adds new book's information into the database
Update Book's Information	H	TID 3.2	Updates book's information from the database
Deactivate Book's Record	H	TID 3.3	Updates book's status to mark it as deactivated record from database
Book Issuance	H	TID 4.0	Processes item issuance
Return Book	H	TID 5.0	Processes returned items
Report			
Transaction Report	M	TID 6.1	Generates specified borrowing transactions
Library Visitors Report	M	TID 6.2	Generates specified library visitors
Borrowers Report	M	TID 6.3	Generates specified borrowers information
Items Report	M	TID 6.4	Generates specified item information
Inventory	M	TID 6.5	Generates item information with quantity

Table 3.0 Features to be Tested

2.2. Features Not to be Tested

Features	Description
Access Role and Rights Management	Software testing is to be done only for library personnel, faculty and staff. This module is intended only for the system's administrator.
Book Classification Management	Software testing is to be done only for library personnel, faculty and staff. This module is intended only for the system's administrator.
User Account Management	Software testing is to be done only for library personnel, faculty and staff. This module is intended only for the system's administrator.
Dashboard	Doesn't require testing.
Library Visitors Monitoring	Doesn't require testing as there is no interaction to be done in the user interface.

Table 4.0 Features Not to be Tested

2.3. Testing Tools and Environment

The Library Automate testing is consisting of Unit, System, Integration and Acceptance Test levels. Software testing is to be done by the school librarian and staff as audience with the assistance of software developer. Furthermore, the rest of the test will be done by the software developer with the guidance of the project adviser. The template that will be used for designing test case is shown in the figure below.

Test Case ID		Test Case Description			
Created By		Reviewed By		Version	
QA Tester's Log					
Tester's Name			Date Tested		Test Case
S#	Prerequisites		S#	Test Data Requirement	
1			1		
2			2		
3			3		
4			4		
Test Conditions					
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	

Table 5.0 Test Case Template

3. Test Cases

3.1. Login Test Case

Test Case ID	TID 1.1	Test Case Description	Test Case for Librarian Login		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Username = MariaNila.Mejorada		
2	Compatible web browser	2	Password = Mejorada123456		
Test Conditions	Valid user credentials				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Open web browser and type "localhost" in URL section	Web system should open	Login page appears	Pass	
2	Enter user credential	None	None	Pass	
3	Click login	Should redirect to dashboard page	Dashboard page appears	Pass	

Table 6.0 Login Test Case: Librarian

Software Test Document
Library Automate

Test Case ID	TID 1.2	Test Case Description	Test Case for Library Staff Login		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Flavia Ong	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Username = Flavia.Ong		
2	Compatible web browser	2	Password = Ong123456		
Test Conditions	Valid user credentials				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Open web browser and type "localhost" in URL section	Web system should open	Login page appears	Pass	
2	Enter user credential	None	None	Pass	
3	Click login	Should redirect to dashboard page	Dashboard page appears	Pass	

Table 7.0 Login Test Case: Library Staff

Software Test Document
Library Automate

Test Case ID	TID 1.3	Test Case Description	Test Case for Faculty Login		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Lordwin A. Crescencio	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Username = Lordwin.Crescencio		
2	Compatible web browser	2	Password = Crescencio123456		
Test Conditions	Valid user credentials				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Open web browser and type "localhost" in URL section	Web system should open	Login page appears	Pass	
2	Enter user credential	None	None	Pass	
3	Click login	Should redirect to dashboard page	Dashboard page appears	Pass	

Table 8.0 Login Test Case: Faculty

Software Test Document
Library Automate

Test Case ID	TID 1.4	Test Case Description	Test Case for Student Login		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Lester John Salon	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Username = LesterJohn.Salon		
2	Compatible web browser	2	Password = Salon123456		
Test Conditions	Valid user credentials				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Open web browser and type "localhost" in URL section	Web system should open	Login page appears	Pass	
2	Enter user credential	None	None	Pass	
3	Click login	Should redirect to dashboard page	Dashboard page appears	Pass	

Table 9.0 Login Test Case: Student

3.2. Borrowers Information Management Test Case

Test Case ID	TID 2.1	Test Case Description	Test Case for Add Borrower's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Borrower Type = Student		
2	Compatible web browser	2	First Name = Mathew		
3	Logged in user	3	Middle Name = Mahumot		
		4	Last Name = Dumape		
		5	Suffix = None		
		6	Address = Mahayag, San Miguel, Bohol		
		7	Contact Number = None		
Test Conditions	Borrower's fullname must not exist from database. If borrower type is student, the user must enter student's id number.				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click add button	A form dialog box should appear	Form dialog box appears	Pass	
2	Fill-up necessary information	Mandatory fields should be filled-in	Mandatory fields are filled-in	Pass	
3	Click submit	Should prompt user that information has been saved	Prompts user that information has been saved	Pass	

Table 10.0 Borrowers Information Management Test Case: Add Borrower's Information

Software Test Document
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Test Case ID	TID 2.2	Test Case Description	Test Case for Update Borrower's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Borrower Type = Student		
2	Compatible web browser	2	First Name = Mathew		
3	Logged in user	3	Middle Name = Mahumot		
		4	Last Name = Dumape		
		5	Suffix = None		
		6	Address = Mahayag, San Miguel, Bohol		
		7	Contact Number = None		
Test Conditions	Borrower's fullname must not exist from database. If borrower type is student, the user must enter student's id number.				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click update button	A form dialog box should appear	Form dialog box appears	Pass	
2	Update fields need to modify	Mandatory fields should be filled-in	Mandatory fields are filled-in	Pass	
3	Click submit	Should prompt user that information has been updated	Prompts user that information has been updated	Pass	

Table 11.0 Borrowers Information Management Test Case: Update Borrower's Information

Software Test Document
Library Automate

Test Case ID	TID 2.3	Test Case Description	Test Case for Deactivate Borrower's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed				
2	Compatible web browser				
3	Logged in user				
Test Conditions	None				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click remove button	A confirmation dialog box should appear	Confirmation dialog box appears	Pass	
3	Click OK	Should prompt user that information has been removed	Prompts user that information has been removed	Pass	

Table 12.0 Borrowers Information Management Test Case: Deactivate Borrower's Information

3.3. Books Information Management Test Case

Test Case ID	TID 3.1	Test Case Description	Test Case for Add Book's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Location = TMC / Public Library		
2	Compatible web browser	2	Accession Number = TMC-00000001		
3	Logged in user	3	ISBN = 19310919		
		4	Requesting Department = College of Computer Studies		
		5	Book Title = Introduction to Theory of Computation		
		6	Author = Michael Sipser		
		7	Category = Book		
		8	Book Classification = 600		
		9	Publisher = CENgage Learning		
		10	Cover = Hard Bound		
		11	Number of Pages = 861		
		12	Volume = 0		
		13	Edition = Third Edition		
		14	Barcode = 123909190		
		15	Receive Date = 2019-10-01		
		16	Acquisition Type = New		
		17	Max No. of Days to Borrow = 7		
		18	Remarks = None		

Test Conditions		Mandatory fields must be filled-in.		
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Click add button	A form dialog box should appear	Form dialog box appears	Pass
2	Fill-up necessary information	Mandatory fields should be filled-in	Mandatory fields are filled-in	Pass
3	Click submit	Should prompt user that information has been saved	Prompts user that information has been saved	Pass

Table 13.0 Books Information Management Test Case: Add Book's Information

Software Test Document
Library Automate

Test Case ID	TID 3.2	Test Case Description	Test Case for Update Book's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Location = TMC / Public Library		
2	Compatible web browser	2	Accession Number = TMC-00000001		
3	Logged in user	3	ISBN = 19310919		
		4	Requesting Department = College of Computer Studies		
		5	Book Title = Introduction to Theory of Computation		
		6	Author = Michael Sipser		
		7	Category = Book		
		8	Book Classification = 600		
		9	Publisher = CENgage Learning		
		10	Cover = Hard Bound		
		11	Number of Pages = 861		
		12	Volume = 0		
		13	Edition = Third Edition		
		14	Barcode = 123909190		
		15	Receive Date = 2019-10-01		
		16	Acquisition Type = New		
		17	Max No. of Days to Borrow = 7		
		18	Remarks = None		
Test Conditions	Mandatory fields must be filled-in.				

Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended
1	Click update button	A form dialog box should appear	Form dialog box appears	Pass
2	Update fields need to modify	Mandatory fields should be filled-in	Mandatory fields are filled-in	Pass
3	Click submit	Should prompt user that information has been updated	Prompts user that information has been updated	Pass

Table 14.0 Books Information Management Test Case: Update Book's Information

Test Case ID	TID 3.3	Test Case Description	Test Case for Deactivate Book's Info		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Reason for removing = Obsolete / Old / Lost		
2	Compatible web browser				
3	Logged in user				
Test Conditions		None			
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click remove button	A confirmation dialog box should appear	Confirmation dialog box appears	Pass	
3	Click OK	Should prompt user that information has been removed	Prompts user that information has been removed	Pass	

Table 15.0 Books Information Management Test Case: Deactivate Book's Information

3.4. Book Issuance Test Case

Test Case ID	TID 4.0	Test Case Description	Test Case for Book Issuance		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Transaction Number = TR00000001		
2	Compatible web browser	2	Borrower ID = S-00000001		
3	Logged in user	3	Book ID = 1		
Test Conditions	Borrower's information must be supplied. At least one book must be present in the cart.				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click search borrower	List of registered borrowers must be displayed in a dialog box	List of registered borrowers are displayed in a dialog box	Pass	
2	Select borrower info	Borrower's info must be transferred to book issuance's page	Borrower's information is displayed	Pass	
3	Click search books	List of books must be displayed in a dialog box	List of book are displayed in a dialog box	Pass	
4	Click submit	Should prompt user that transaction has been processed	Prompts user that transaction has been processed	Pass	

Table 16.0 Book Issuance Test Case

3.5. Return Book Test Case

Test Case ID	TID 5.0	Test Case Description	Test Case for Return Book		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Transaction Number = TR00000001		
2	Compatible web browser				
3	Logged in user				
Test Conditions		Must select books to be returned from the cart.			
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click search borrowing transactions	List of borrowing transactions must be displayed in a dialog box	List of borrowing transactions are displayed in a dialog box	Pass	
2	Select books to returned	Selected books must have a checked status in its check box	Selected books has checked status in its check box	Pass	
3	Click submit	Should prompt user that transaction has been processed	Prompts user that transaction has been processed	Pass	

Table 17.0 Return Book Test Case

3.6. Reports Test Case

Test Case ID	TID 6.1	Test Case Description	Test Case for Transaction Reports		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Book Status = Unreturned		
2	Compatible web browser	2	Status = Active		
3	Logged in user				
Test Conditions	None				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Select filtering options	None	None	Pass	
2	Click filter	Must display all unreturned books.	Displays all unreturned books	Pass	

Table 18.0 Reports: Transactions

Software Test Document
Library Automate

Test Case ID	TID 6.2	Test Case Description	Test Case for Library Visitors Reports		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Fiscal Year Start = 2019		
2	Compatible web browser	2	Fiscal Year End = 2020		
3	Logged in user	3	Semester = 1		
		4	Status = ""		
Test Conditions	None				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Select filtering options	None	None	Pass	
2	Click filter	Must display all library visitors from fiscal year 2019-2020 first semester.	Displays all library visitors from fiscal year 2019-2020 first semester.	Pass	

Table 19.0 Reports: Library Visitors

Test Case ID	TID 6.3	Test Case Description	Test Case for Borrowers Reports		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Option = Most Issuing Borrower		
2	Compatible web browser				
3	Logged in user				
Test Conditions	None				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Select filtering options	None	None	Pass	
2	Click filter	Must display all borrowers with the most issued items	Displays display all borrowers with the most issued items	Pass	

Table 20.0 Reports: Borrowers

Test Case ID	TID 6.4	Test Case Description	Test Case for Items Reports		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites	S#	Test Data Requirement		
1	Web system deployed	1	Option = Most Issued Item		
2	Compatible web browser				
3	Logged in user				
Test Conditions	None				
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Select filtering options	None	None	Pass	
2	Click filter	Must display all items with the most number of issues	Displays all items with the most number of issues	Pass	

Table 21.0 Reports: Items

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Test Case ID	TID 6.5	Test Case Description	Test Case for Inventory Reports		
Created By	Clark Kevin Villamor	Reviewed By	Jasmine A. Tulin	Version	1.0
QA Tester's Log					
Tester's Name	Maria Nila A. Mejorada	Date Tested	October 8, 2019	Test Case	Pass
S#	Prerequisites		S#	Test Data Requirement	
1	Web system deployed				
2	Compatible web browser				
3	Logged in user				
Test Conditions		None			
Step #	Step Details	Expected Results	Actual Results	Pass / Fail / Not Executed / Suspended	
1	Click Inventory Tab	Must display all items with its total quantity and total quantity on hand	Displays all items with its total quantity and total quantity on hand	Pass	

Table 22.0 Reports: Inventory